RR-F-00425(GSA-FSS)
November 1, 1955

INTERIM FEDERAL SPECIFICATION

RR-F-00425(GSA-FSS)

FLARES, MARKER, GROUND OBSTRUCTION

(LIQUID-BURNING, EMERGENCY)

This Interim Federal Specification is based upon currently available technical information, but has not been approved for promulgation as a regular Federal specification. It is subject to modification, and pending its promulgation as a regular Federal specification, it is for optional use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - The ground obstruction marker flares covered by this specification, display a warning light, indicating to the driver of an approaching vehicle that beyond the burning flare there is a hazard and he should proceed with caution. The fuel used in the flare is kerosene or other liquid fuel having a fire hazard classification not greater than kerosene.

1.1.1 Federal specification coverage. - Federal specifications do not cover all types, classes and styles of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. - The flares covered by this specification shall be of the following types, classes, styles, and duties, as specified (see 6.1):

Type I. - Set of three, for highway automotive use.
  Class A. - Flat bottom, nonself-righting.
    Style 1. - In metal box (fig. 1).
      12-hour duty. - Minimum of 12 hours burning time per flare.
    Style 2. - In metal racks (fig. 2).
      12-hour duty. - Minimum of 12 hours burning time per flare.

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Type II. - Set of three, for highway-construction use.
Class A. - Flat bottom, nonself-righting (fig. 3).
24-hour duty. - Minimum of 24 hours burning time per flare.
36-hour duty. - Minimum of 36 hours burning time per flare.

Class B. - Round bottom, self-righting (fig. 4).
24-hour duty. - Minimum of 24 hours burning time per flare.
36-hour duty. - Minimum of 36 hours burning time per flare.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications. - The following specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

Federal Specification:
TT-P-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. - The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Governmental:
National Bureau of Standards:
Commercial Standard CS88-41

Interstate Commerce Commission:
Code of Federal Regulations, Title 49 Transportation,
Part 164 to end.
Part 149.8 - Emergency Parts and Accessories Required (Page 223 of 1949 Edition.)

(Copies of the National Bureau of Standards Commercial Standard CS88-41 and the Code of Federal Regulations may be obtained upon application accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

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Nongovernmental:

Society of Automotive Engineers' Standard:
Liquid Burning Emergency Flares
(Report of Lighting Technical Committee.)

Underwriters' Laboratories, Inc. Standard:
Liquid Burning Flares, Subject 912.
Highway Emergency Signals 380 - 19.81.
Signals and Signaling Appliances 380.19.

American Society for Testing Materials Standard:
B 117 - 49T - Salt Spray (Fog) Testing.

(Copies of the SAE Standard may be obtained from the Society of Automotive Engineers, Inc., 29 West 39 Street, New York 18, N. Y.

(Copies of the Underwriters' Laboratories, Inc., standards may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; or 500 Sansome Street, San Francisco 11, Calif.

(Copies of the ASTM Standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Illustrations. - The illustrations shown are for the convenience of identification and are not intended to preclude the purchase of flares which are otherwise in accordance with the requirements of this specification (see 4.1.4.2).

3.2 Design and construction. - The flares shall be designed for use with kerosene or other liquid fuel having a fire hazard not greater than kerosene. Joints, if any, shall be constructed in a manner to prevent leakage. The burner element shall be securely fastened in its proper position. Each type I flare shall be provided with a sealing cap secured so that leakage of fuel will not occur while the flare is not in use. When specified, type II flares shall be furnished with a sealing cap to prevent leakage of fuel while the flare is not in use (see 6.1). The flares when lighted shall be discernable at a distance of 500 feet under normal atmospheric conditions and low visibility. All 12-hour duty flares covered by this specification shall meet the applicable test requirements of section 4, which are essentially the same as the test requirements for liquid burning emergency flares in the Society of Automotive Engineers' Handbook; the longer duty flares shall meet the longer minimum burning time requirements as outlined in section 4.

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3.3 Marking. - Flares shall be marked in a plain and permanent manner with the manufacturer's name or trade-mark of such known character that the source of manufacture may be readily determined.

3.4 Finish. - Unless otherwise specified, the color of the flares shall be the manufacturer's standard (see 6.1). The finish shall withstand the corrosion test requirements of section 4.

3.5 Duty. - Each flare covered by this specification, when filled with kerosene and the wick lighted, shall burn at least the minimum number of hours specified, when tested in accordance with section 4.

3.6 Type I. - Type I flares shall be furnished in sets of three, mounted in either a metal box (see 3.6.2.1) or metal rack (see 3.6.2.2) dependent upon the style specified. The metal box or rack shall be provided with a means whereby it can be padlocked although the padlock need not be furnished. When specified, each type I set of 3 flares shall be furnished with a set of 3 red flags and 3 flagstaffs (see fig. 1 and 6.1).

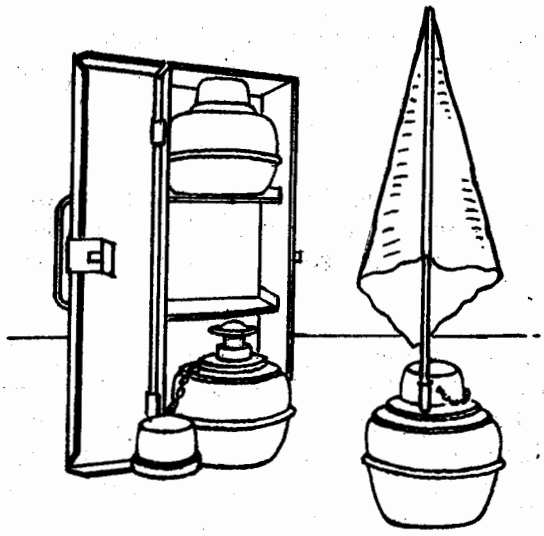


Figure 1. - Type I, class A, style 1 flares, set of 3 for automotive use, flat bottom, nonself-righting, in metal box.

3.6.1 Flags. - The flags when furnished shall be at least 12 x 12 inches square. They shall be of fast color red cotton cloth, the edges of which shall be hemmed or otherwise finished to prevent raveling. The flagstaff shall be either of wood or metal. Each flagstaff shall be at least 15 inches in length and shall fit snugly into the socket on the flare.

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3.6.2 Class A. - Type I, class A flares shall have a flat bottom that is not weighted. For the purpose of this specification, a flat bottom flare shall include any flare having either a completely flat bottom or a bottom which comes in contact with a flat surface, at or nearly to the outer periphery of the flare.

3.6.2.1 Style 1. - Each set of three, type I, class A, style 1 flares shall be furnished in a substantial metal box suitable for mounting on a truck. When flags are specified, the metal box shall be of such size as to retain the 3 flares, 3 flags, and 3 flagstaffs.

3.6.2.2 Style 2. - Each set of three, type I, class A, style 2 flares shall be furnished on a substantial metal rack, suitable for mounting on a truck (see figure 2). The rack shall provide for ease in mounting and removing flares. When flags are specified, a metal container shall be attached to the rack for carrying the 3 flags and 3 flagstaffs.

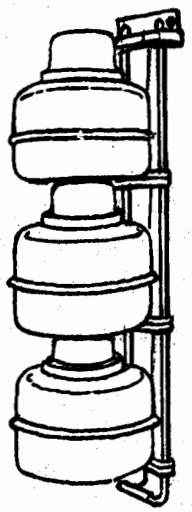


Figure 2. - Type I, class A, style 2 flares, set of 3 for automotive use, flat bottom, nonself-righting, on metal rack.

3.7 Type II. - Type II flares shall be suitable for highway construction use. The body of the flare shall be of molded or pressed steel not less than 0.035 inch in wall thickness. The flares shall be furnished in sets of three. Unless otherwise specified, each set of three flares shall be furnished in a commercial shipping carton conforming to Interstate Commerce Commercial regulations (see 2.2). Each type II flare shall be furnished with a wick at least 3/4 inch in diameter and at least 18 inches long.

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3.7.1 Class A. - Type II, class A, flat bottom flares shall conform to 3.6.2 (see figure 3).

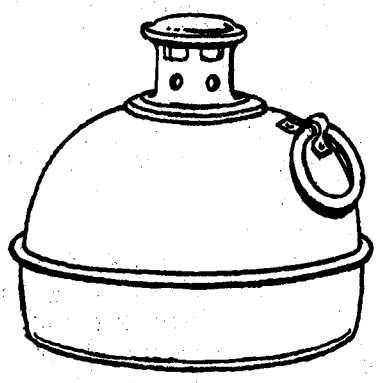


Figure 3. - Type II, class A flares, set of 3 for highway construction use, flat bottom, nonself-righting, (Note: Only 1 flare shown.)

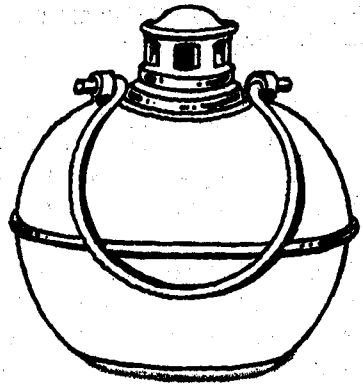


Figure 4. - Type II, class B flares, set of 3 for highway construction use, round bottom, self-righting. (Note: Only 1 flare shown.)

3.7.2 Class B. - Type II, class B, flares shall be provided with a weighted and sufficiently rounded or spherical bottom to return the flare to an upright position in the event it is tipped over (see figure 4).

4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 Type I flare tests. - The following sampling, inspection and test procedures in the subparagraphs of 4.1 are applicable to type I flares. The subparagraphs of 4.1 are similar to the test requirements of Commercial Standard CS-88-41 with certain technical changes to conform to SAE Standard. The requirements from the SAE Handbook were extracted from the following report: Report of Lighting Technical Committee approved January 1949 and last revised January 1953.

4.1.1 Instructions for use. - The manufacturer shall furnish with each set of flares submitted for laboratory test, printed instructions as to wick adjustment, maximum filling level, the method of installation, and such other details as the manufacturer may deem necessary in order that the user of these flares may be able to operate them at their maximum efficiency.

4.1.2 Samples for test. - Sample flares submitted for laboratory test shall be representative of the devices as regularly manufactured and marketed. The various tests may be made on separate samples, if desired, to expedite completion of tests.

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4.1.3 Laboratory facilities. - All laboratory tests shall be made by a recognized impartial engineering laboratory having all facilities and equipment necessary to make accurate physical and optical tests herein specified in accordance with established laboratory practices.

4.1.4 Vibration and shock test. - Three type I sample flares, in the metal rack or metal box in which they were submitted to the laboratory positioned as would be normal in service, shall be bolted to the anvil end of the table of the vibration-test machine (figure 5) and vibrated approximately 750 times per minute through a distance of 1/4 inch. The table shall be spring-mounted at one end and fitted with steel calks on the under side of the other end. These calks are to make contact with the steel anvil once during each cycle at the completion of the fall. The rack shall be operated under a spring tension of 60 to 70 pounds. This test shall be continued for one hour. Similar tests shall be conducted on type II flares.

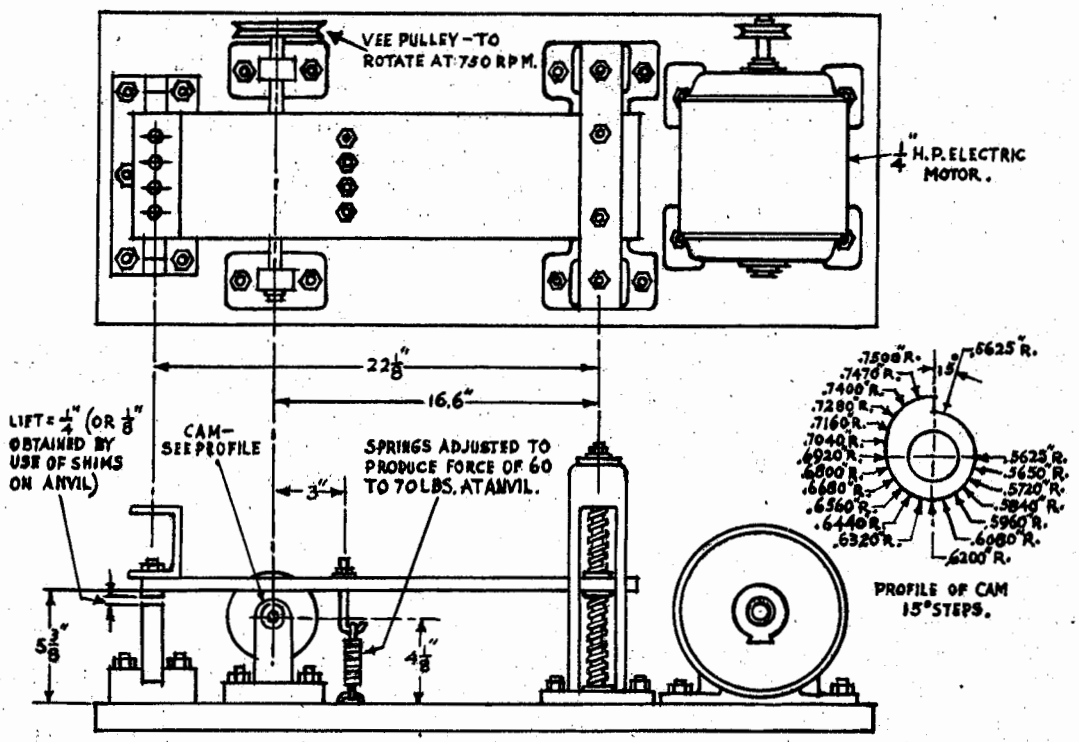


FIGURE 5.- Vibration and shock testing machine.

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4.1.4.1 At completion of the test specified in 4.1.4, the flares shall be examined. Any flare showing evidence of structural failure or leakage in the body or joints at any time during this test fails to meet the requirements of this specification.

4.1.4.2 It is recommended that for the purpose of standardizing the vibration and shock test, it be made on a testing machine substantially in accordance with figure 5.

4.1.5 Corrosion test. - A sample unit shall be subjected to a salt-spray (fog) test in accordance with either the latest ASTM-B-117-49T or Federal Specification TT-P-141 test 606.1 for a period of 50 hours consisting of two periods of 24 hours exposure and one-hour minimum drying time each. There shall be no evidence of excessive corrosion immediately after the above test has been completed, which would affect the proper functioning of the flare.

4.1.6 Weatherproof test. - The flares used for these tests shall be filled with kerosene to the level recommended in the manufacturer's instructions and shall have the wicks adjusted as recommended by the manufacturer.

4.1.6.1 After a preheating period of 5 minutes in still air, a sample flare, lighted and mounted in its normal operating position on a table rotating at 4 revolutions per minute and in a wind of approximately 2 miles per hour, shall be subjected to a water spray from an adjustable, solid-cone nozzle (such as the ordinary garden-hose spray nozzle), set so that the nozzle outlet is 8 to 10 feet horizontally from the sample and 1 to 3 feet vertically above the sample, with the nozzle axis pointing upward at an angle of approximately 45 degrees with the horizontal in a downward direction.

4.1.6.2 Under the conditions specified above and with a water pressure of 5 to 7½ pounds per square inch at the nozzle, the rate of precipitation at the location of the sample shall be adjusted to 0.10 inch per minute. The sample flare shall be introduced gradually into the spray and after being placed in the test location shall continue to operate under these conditions for 15 minutes. This test shall be made on each of the three flares constituting a set, and at least two out of three flares shall pass the test.

4.1.6.3 With the rate of rotation and the wind condition the same as described above, the water pressure shall be further increased to a value of 18 to 20 p.s.i. and the sample moved farther from the nozzle, if necessary, to a location giving a precipitation of 0.01 inch per minute, striking the sample at approximately 45 degrees with the horizontal. Under these conditions, the sample shall continue to operate for 1 hour and 30 minutes. This test shall be made on each of the three flares constituting a set, and at least two out of three flares shall pass the test.

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4.1.7 Reliability, stability, and life tests. - The flares used for these tests shall be filled with kerosene to the level recommended by the manufacturer and shall have the wicks adjusted in accordance with the manufacturer's instructions.

4.1.7.1 A flare that is nearly empty of fuel shall be lighted, placed on a horizontal concrete surface similar to a pavement, and allowed to burn in still air for a preheating period of 5 minutes. It shall then be placed suddenly in an air stream of 40 miles per hour and withdrawn. After rotating through approximately 45 degrees it shall again be placed in the air stream and withdrawn. It shall be rotated again through an additional 45 degrees approximately and placed in the air stream a third time. If the flame is extinguished or if the flare slides or is overturned by the wind on any one of these three operations, the sample shall have failed to pass the test. The test shall then be repeated on each of the other two samples constituting a set and at least two out of three flares shall pass the test.

4.1.7.2 A flare, lighted and mounted in its normal operating position, rotating about its vertical axis at 4 revolutions per minute, shall be subjected to a horizontal current of air having a velocity the equivalent of wind at 40 m.p.h. This test shall continue for 15 minutes and the flare shall remain lighted throughout the entire 15 minute period.

4.1.7.3 Upon completion of the above test the flare shall, while lighted and rotated as specified above, be subjected to a horizontal current of air having a velocity of 5 m.p.h. The total interrupted burning time, including the first 15 minutes at 40 m.p.h., shall be at least 12 hours. The flare shall not be refilled during tests; there shall be no adjustment of wick.

4.1.7.4 The flare shall operate satisfactorily for a period of at least 2 hours in still air.

4.1.7.5 Photometric test. - A sample flare, when subjected to a wind velocity of 5 m.p.h. and 40 m.p.h., respectively, shall produce a minimum of 0.10 candle power in a horizontal direction.

4.2 Type II flare tests. - Type II flares shall be subjected to the same test requirements as type I flares except for the substitution of the drop test for the vibration and shock test and the longer minimum burning time test (see 4.2.1 and 4.2.2).

4.2.1 Drop test. - In lieu of the vibration and shock test (4.1.4 to 4.1.4.2 inclusive) for type I flares, the type II flares shall be subjected to a drop test. Each empty flare shall be dropped on a concrete pavement 5 times from a height of 4 feet with the seam (if any) side down. The type II, class B flares with weighted bottoms may be dropped, two tied together, so as to get the proper balance for dropping them seam side down. After the drop test, each sample flare shall be refilled with kerosene and allowed to stand at least one hour, the flares shall then be examined and shall show no leakage. This test shall be made on each of the three flares constituting a set and all three flares shall pass the test.

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4.2.2 Uninterrupted burning time test. - In lieu of the minimum uninterrupted burning time of 12 hours as required of type I flares (see 4.1.7.3), the type II flares when tested under the same conditions as type I flares, shall have a minimum uninterrupted burning time as the duty specified, the 24 hour duty flares shall burn at least 24 hours and the 36 hour duty flares shall burn at least 36 hours.

5. PACKAGING FOR DELIVERY

5.1 Packaging. - Unless otherwise specified, commercial packaging is acceptable under this specification (see 3.6 and 6.1).

5.2 Packing. - Unless otherwise specified, the subject commodity shall be delivered in commercial containers so constructed as to insure safe delivery by common or other carrier to the point of delivery (see 6.1).

5.3 Marking. - Unless otherwise specified, shipping containers shall be marked with the name of the article, the type, class, style, if any, duty, and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor and the number of the contract or order (see 6.1).

6. NOTES

6.1 Ordering data. - Purchasers should specify the type, class, style if any, and the duty, of the flares required (see 1.2). Purchasers should exercise any of the desired options offered herein:

(a) If sealing caps are required on type II flares, so state (see 3.2).
(b) Color of flares, if different (see 3.4).
(c) If red flags are required for type I flares, so state (see 3.6).
(d) If type II flares are to be furnished in any other container than the shipping carton, then specify the kind of box or other container required (see 3.7).
(e) Packaging, packing, and marking requirements, if different (see section 5).

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6.2 Compliance with existing standards, reports, and regulations. - It is intended that the flares covered by this specification meet at least the mimimum requirements of existing standards, reports, and regulations covering liquid burning flares except that special marking requirements beyond the manufacturer's identification are not a requirement of this specification. Therefore any flare covered by this specification is intended to meet the applicable minimum (except special marking) requirements of all of the documents listed herein, but flares meeting the requirements of some of the listed documents will not necessarily meet the requirements of this specification. Standards, reports, and regulations forming a basis of some portions of this specification are as follows (see 2.1 and 2.2):

National Bureau of Standards' Commercial Standard CS-88-41 for Liquid Burning Flares.

SAE Standard for Liquid Burning Emergency Flares (Report of Lighting Technical Committee).

Underwriters' Laboratories, Inc. Standard for Liquid Burning Flares, Subject 912, Highway Emergency Signals 380 19.81 and Signals and Signaling Appliances 380.19.

Code of Federal Regulations, Title 49 - Transportation, Part 164 to end, and Part 149.8 - Emergency Parts and Accessories Required - Page 223 of 1949 Edition

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